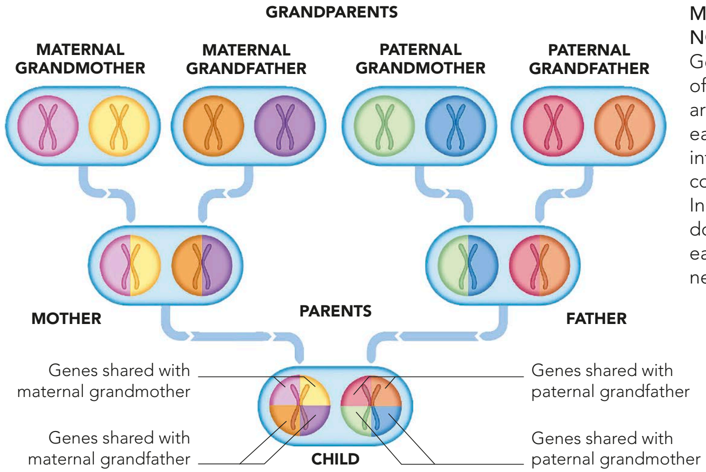
GENERATIONAL SEQUENCE

The two versions of the genes (alleles) are mixed, or reshuffled, as they are inherited at each generation. In effect, a child inherits one-quarter of its total genes from each grandparent. The child's inherited features strongly resemble a mixture of those from his or her parents, but the features from the grandparents appear to be less marked.

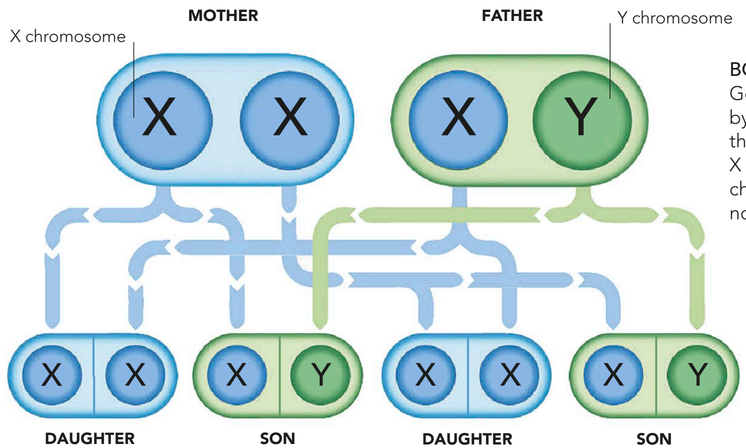


MIXED, BUT NOT BLENDED

Genes are "units" of inheritance that are shuffled at each generation into different combinations. Individual genes do not blend with each other to create new versions.

INHERITANCE OF GENDER

Gender depends on which sex chromosome – an X or a Y – is inherited. Females have two identical Xs; males have an X and a smaller Y, with male genes. A woman's egg cells all contain an X, whereas half a man's sperm cells contain an X and the other half a Y. The gender of offspring is always determined by the father.



BOY OR GIRL?

Gender is determined by the inheritance of the sex chromosomes, X and Y (the other 22 chromosome pairs are not shown here).